



Disassembly Strategy Can Improve Engine Turnaround Time

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

One method that can have a great impact on improving engine turnaround time (TAT) is removing sub-assembly groups as efficiently as possible, and further disassembling/inspecting these groups in other dedicated bays outside the main disassembly bay. This is in an effort to get to the bare block, on which all assembly activities are dependent on its readiness. Since there is detailed cleaning, inspection, salvage work, and possible logistics issues (if a new or Reman block is required), getting the engine disassembled quickly can have a major impact on overall TAT.

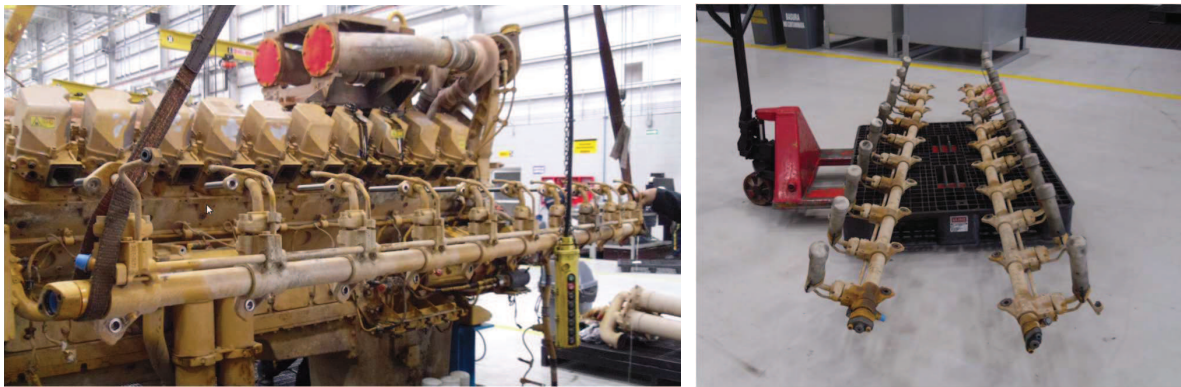
2.0 Best Practice Description

At Matco CRC in Hermosillo, Sonora MX, engines are disassembled in one of three main disassembly bays by removing sub-assembly groups, which are then moved to three subassembly disassembly bays across the aisle where they are further broken down. With 2-3 technicians working a combined ~50 man hours (including breaks), Matco can have a before-failure C175-20 fully disassembled down to the bare block. The following photos illustrate their disassembly strategy, which is to remove sub-assembly groups wherever possible, for further disassembly outside the main disassembly bay.

Oil cooler assembly



Fuel rails with quill tubes attached



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Oil filter housing



Oil pan assembly

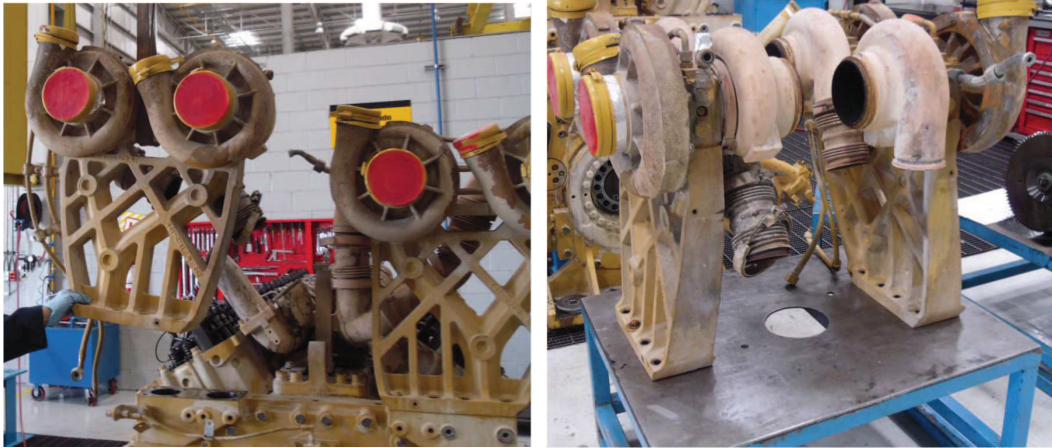


Turbo step platform

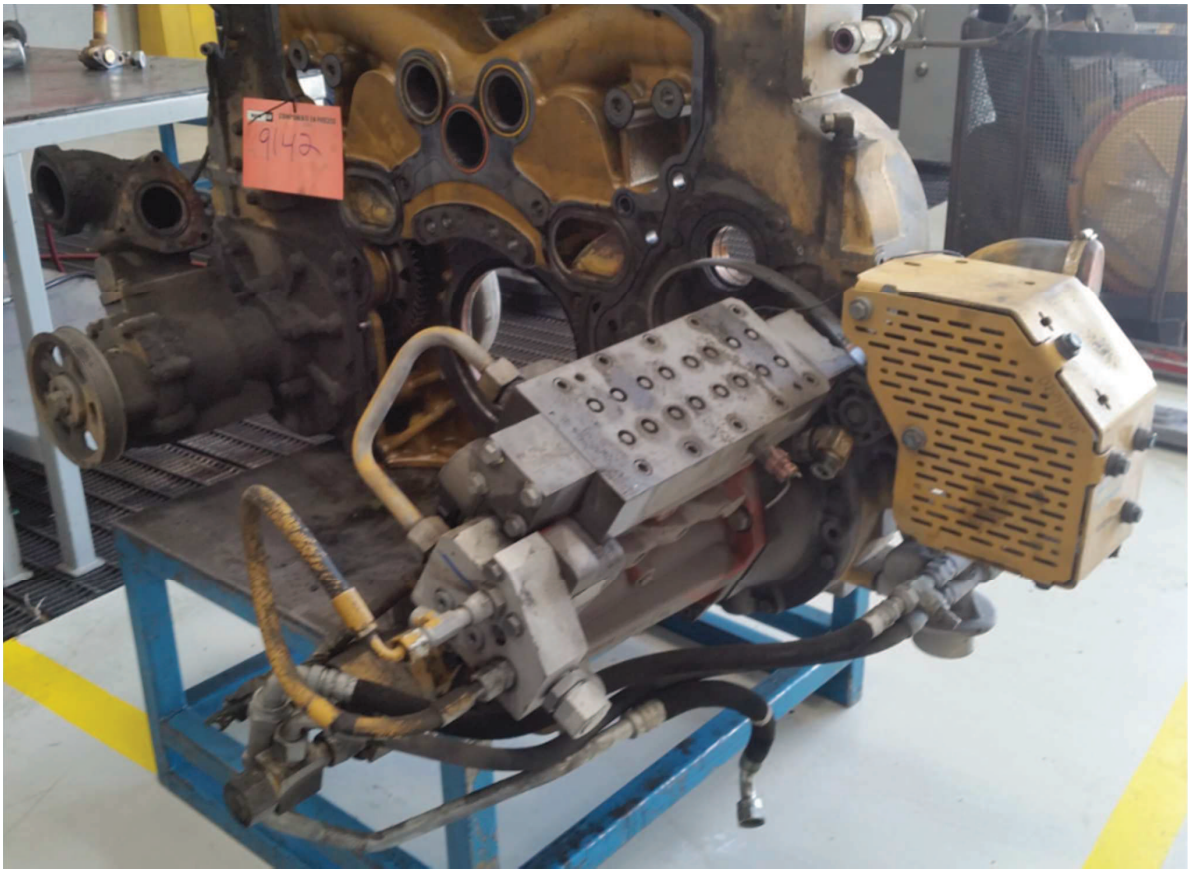


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Turbos and turbo supports



Front housing assembly w/ HPFP (assembly with tubes, monoblock, & transfer pump), oil pump, & water pumps

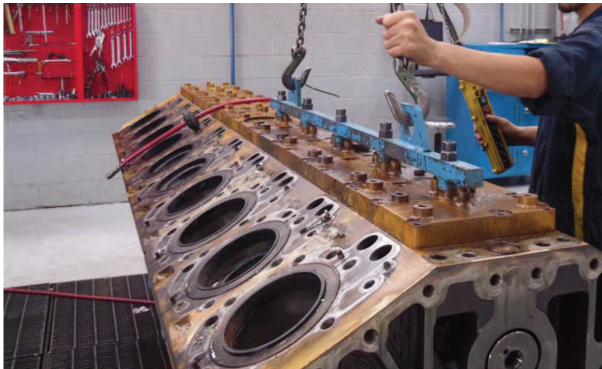


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Heads with exhaust manifolds in place



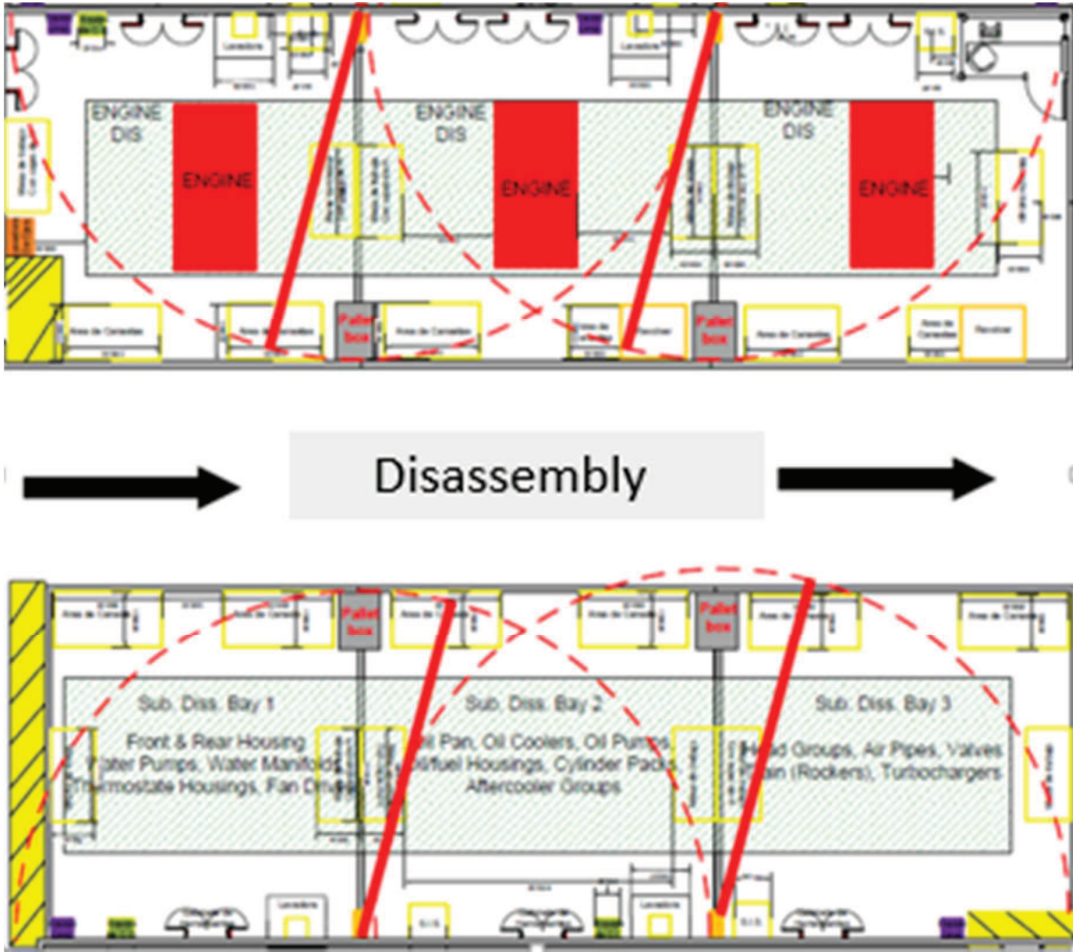
Lifter groups



Other items not shown here that are moved to the sub-component bays to have further disassembly/inspection work done include:

- Air intake manifolds
- Oil pressure relief valve
- Fuel crossover tubes
- ECM (removed from bracket, and ET download)
- Damper
- Fuel filter base
- Oil pump & scavenge pump
- Rear housing

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3.0 Implementation Steps

- CRC layout or bay assignment changes
- Process maps, checklists, standard work updated
- Assess additional manpower requirements
- Monitor and adjust new processes for maximum efficiency

4.0 Benefits

A more efficient overall disassembly process with specialized technicians, which can enable a bare block to be obtained quickly, therefore improving overall engine TAT.

5.0 Resources Required

- Technicians in sub-component disassembly bays
- Clerical (Process maps, checklists)
- Additional sub-assembly fixtures/stands, and lifting devices as required

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6.0 Supporting Attachments / References

7.0 Related Best Practices

8.0 Acknowledgements

Disassembly process documented and photographed during engine rebuild at MATCO CRC in Hermosillo, Sonora MX

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